

Solve and explain equations as balanced relationships

Name: _____

Date: _____

Step 9

Middle School



Today I am learning to solve equations and explain how they show balanced relationships.



1. Balanced or Not?

Look at each equation. Circle.

a) $5 + 3 = 8$

☐ Balanced ☐ Not Balanced

b) $7 + 2 = 10$

☐ Balanced ☐ Not Balanced

c) $12 - 4 = 8$

☐ Balanced ☐ Not Balanced

d) $6 + 5 = 12$

☐ Balanced ☐ Not Balanced

2. Solve the Unknown

Find the missing value.

a) $x + 4 = 10$ $x =$ _____

b) $n - 3 = 5$ $n =$ _____

c) $m + 7 = 15$ $m =$ _____

d) $p - 8 = 12$ $p =$ _____

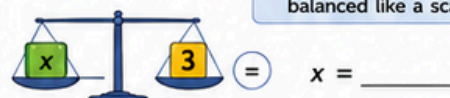
e) $a + 12 = 20$ $a =$ _____

f) $b - 5 = 9$ $b =$ _____

3. Think Like a Balance

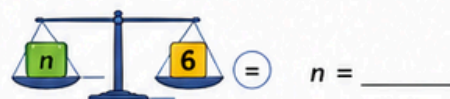
Use the balance idea.

a) $x + 3 = 9$



An equation is balanced like a scale.

b) $n + 6 = 14$



c) $m - 4 = 7$



4. Explain Your Thinking

a) $x + 5 = 13$

How did you find x?

b) $n - 7 = 9$

Explain your method.

5. Match the Equation

Draw a line to match.

1 A number plus 3 equals 10

2 A number minus 5 equals 7

3 A number plus 8 equals 20

4 A number minus 2 equals 9

$n + 3 = 10$

$n - 5 = 7$

$n + 8 = 20$

$n - 2 = 9$

6. Real-Life Equations

Write an equation and solve.

a) Ben has some stickers. He gets 4 more. Now he has 12.



Equation: _____

Answer: _____

b) Mia had some marbles. She gave away 6. Now she has 8 left.



Equation: _____

Answer: _____

c) A box had some pencils. 5 were added. Now there are 18.



Equation: _____

Answer: _____

7. Create Your Own Equation

Write a short story.



Write an equation. _____

Solve it. _____

8. Think and Explain



a) What does the equals sign mean?



b) Why is an equation like a balance?



c) How do you know your answer is correct?



I can...

- ☐ solve simple equations.
- ☐ show that equations are balanced.
- ☐ explain my thinking clearly.



My Learning Goal

I can solve and explain equations as balanced relationships.

